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Collaborators:

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**Instructions:** Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

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## Identify Solutions of 2nd-Order ODEs

### 1. Verify a Solution to a 2nd-Order ODE

Similar to verifying a solution to a 1st-order ODE, to verify a solution to 2nd-order ODE means to check if a proposed function satisfies the given ODE.

If an ODE solution is proposed, we verify it using the following process:

- Step 1: Compute the derivatives of the solution.
- Step 2: Substitute the given solution and its derivatives into the ODE.
- Step 3: Simplify and check if the resulting equation is an identity. If it is, the solution is verified.

a. Consider the ODE  $y'' + 4y = 0$ . For the following functions, determine if they are a solution to the given ODE.

i.  $y_1 = \cos(2t)$

ii.  $y_2 = \sin(2t)$

iii.  $y_3 = 0$

vi.  $y_4 = \cos(2t) + \sin(2t)$

b. For each function, determine if they are a solution for the ODE  $y'' - 3y' + 2y = 0$ .

i.  $y_1 = e^{2t}$

ii.  $y_2 = e^t$

A linear 2nd-order ODE can be written in *standard form*

where  $p$ ,  $q$ , and  $f$  are functions of  $t$  or constants.

- Consider the ODE

The proposed solutions to the given ODE are

d. Show that the equilibrium solution is  $y = \frac{1}{2}$ .